

Frozen question authority and help contract

QUESTION CUSTODY

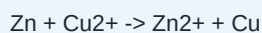
The three questions in this review fixture are owner-frozen inputs. Core2 may add ladder hints and answer closure but may not silently rewrite the question or claim NCERT provenance.

HELP LADDER

Attempt -> small clue -> stronger clue -> full answer/check. The learner decides when to reveal help; the PDF does not detect responses.

RQ-REDOX-01 | Attempt

GOVERNED REACTION



FROZEN QUESTION

For $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$, prove which species is oxidised and which is reduced using oxidation-state changes.

LADDER HINTS

H1: Track the same element. H2: Write oxidation-state lanes. H3: Connect direction to electron loss/gain. H4: Attach agent role only after self-change is proved.

RQ-REDOX-01 | Answer closure

FULL ANSWER

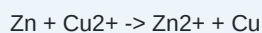
Zn: 0 -> +2, so Zn is oxidised. Cu in Cu²⁺: +2 -> 0, so Cu²⁺ is reduced.

CHECK

Verify element identity, direction of oxidation-state change, two-electron balance, and role-label consistency.

RQ-REDOX-02 | Attempt

GOVERNED REACTION



FROZEN QUESTION

Write the two electron statements and identify the reducing and oxidising agents.

LADDER HINTS

H1: Track the same element. H2: Write oxidation-state lanes. H3: Connect direction to electron loss/gain. H4: Attach agent role only after self-change is proved.

RQ-REDOX-02 | Answer closure

FULL ANSWER

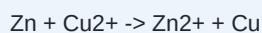
$\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$. $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$. Zn is the reducing agent; Cu^{2+} is the oxidising agent.

CHECK

Verify element identity, direction of oxidation-state change, two-electron balance, and role-label consistency.

RQ-REDOX-03 | Attempt

GOVERNED REACTION



FROZEN QUESTION

A learner says: 'Zn is the reducing agent because Zn gets reduced.' Diagnose the error and repair the reasoning.

LADDER HINTS

H1: Track the same element. H2: Write oxidation-state lanes. H3: Connect direction to electron loss/gain. H4: Attach agent role only after self-change is proved.

RQ-REDOX-03 | Answer closure

FULL ANSWER

The claim reverses Zn's self-change. Zn changes 0 -> +2, so Zn is oxidised; because its electron loss reduces Cu^{2+} , Zn is the reducing agent.

CHECK

Verify element identity, direction of oxidation-state change, two-electron balance, and role-label consistency.